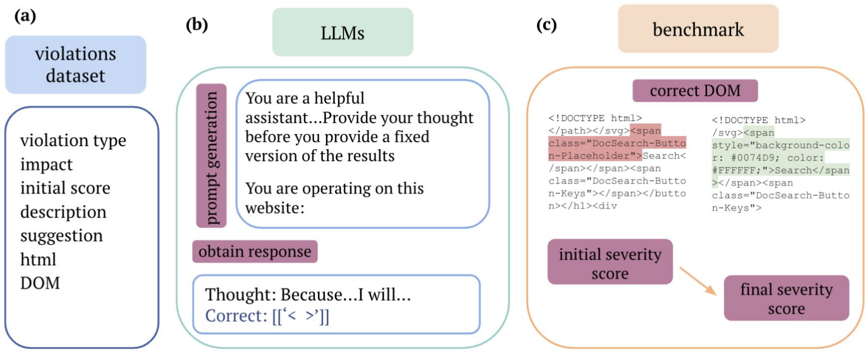
ACCESS Huang Et Al. 2024
ICLR

Problem

- 96%+ of homepages have WCAG violations
- On average ~50 accessibility errors per website
- Lots of tools for detecting violations, not many for actually correcting them → manual correction can take a long time

What they did

- Developed ACCESS Agent:
  - Uses playwright to detect accessibility violations
  - Feeds violations into LLMS (GPT-3.5-turbo-16k and minimal testing with GPT-4)
  - Generates corrected HTML
  - Updates DOM + re-evaluates
- ACCESS Benchmark: gives a score based on severity of violations
  - Cosmetic = 1, Minor = 2, Moderate = 3, Serious = 4, Critical = 5
  - Computes aggregate score + measures % reduced after correction



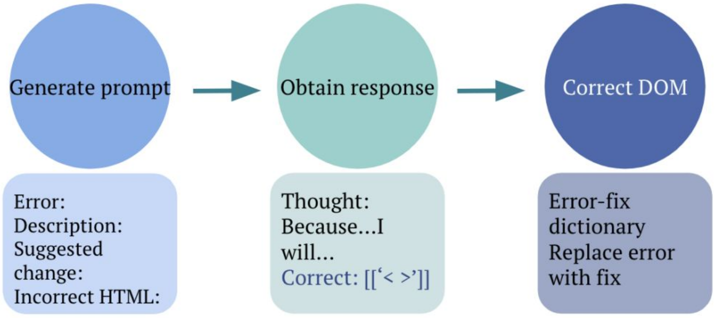


Figure 5: The DOM correction process using prompt engineering and LLMs.

Research Gap

- Previous research focused heavily on detecting violations, rather than correcting
- Previous research mostly focused on Image alt-text generation, instead of other accessibility issues
- No tool to actively modify DOM and directly serve users
- No standardized benchmark to evaluate corrected performance.

Evaluation

Table 2: Initial and Final Violation Severity Score Results

Model	**Prompt**	**Initial / Avg**	**Final / Avg**	**% Score Decrease**
GPT-3.5-turbo-16K	Few-Shot Guidance	614 / 24.56	372 / 14.88	39.414%
GPT-3.5-turbo-16K	Chain of Thought	614 / 24.56	334 / 13.36	45.603%
GPT-3.5-turbo-16K	ReAct	614 / 24.56	299 / 11.96	51.303%

- Text-based violations were corrected more effectively than visual/layout issues.

Future Scope

- Larger & More Diverse Dataset— more geographical/industrial diversity
- Integrate vision models— might help with addressing visual/layout issues
- Better DOM understanding— focuses on one line at a time